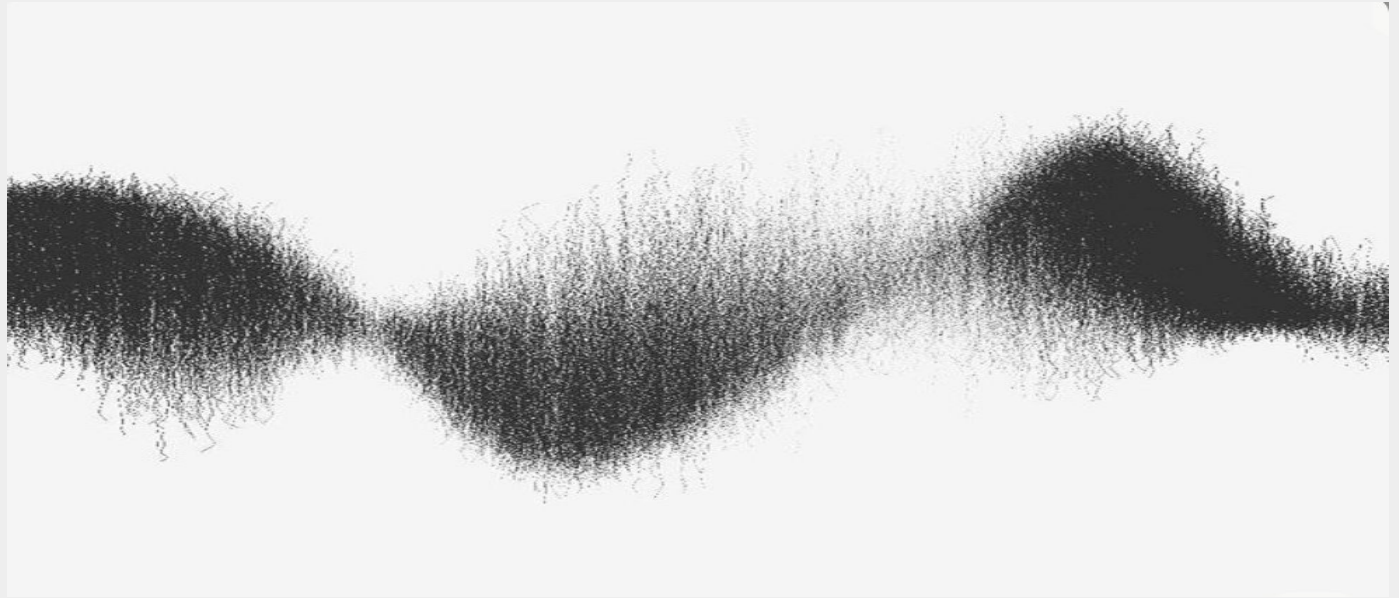


SPOTIFY DATASET ANALYSIS



12-02-2025

CSC-405

GROUP 7

UNCG DEPT. OF
COMPUTER SCIENCE

STAGE I REVIEW: DATASET SELECTION & SETUP

Problem Identification

Our group began by first exploring the Spotify dataset, and concluded from what we found that based on the available features and qualities of the dataset, **an analysis on the likelihood of a song to win a Grammy award, based on certain features**, would be a fitting examination.

An analysis on what makes an award-winning song would be relevant to industry creatives, and interesting to listeners to see what we intuitively look for in music.

STAGE I CONT'D

Dataset Exploration

A standout feature from the beginning for our group was the **popularity** feature, which takes both the individual song and the artist's overall pre-existing popularity into account according to the dataset creator for its calculation.

Feature	Type: Int / float / ...	examples	Missing %
Artist name	String	Jason Mraz, Zack Tabudlo	0%
Track name	String	I'm Yours, Give Me Your Forever	0%
Popularity (chase)	Int	73, 55	
Duration_ms (Christian)	Int	230666, 210826, 198712	0%
Explicit (Hiwete)	Boolean	TRUE, FALSE	0%
Track_genre (Christian)	String	Acoustic, K-Pop	0%
Valence (Isaiah)	Int	.769, .289	0%
Danceability (Hiwete)	Float	.9, .791	0%
Time_signature (chase)	Int	5, 4	0%
Tempo (Isaiah)	Float	87.917, 77.489	0%

Dictionary of our desired dataset features



	track_id	artists	album_nai	track_nam	popularity	duration_r	explicit	danceabili	energy	key	loudness	mode	speechine	acousticn	instrumen	liveness	valence	tempo	time_signe	track_genre
0	5SuOikwif	Gen Hoshi	Comedy	Comedy	73	230666	FALSE	0.676	0.461	1	-6.746		0	0.143	0.0322	1.01E-06	0.358	0.715	87.917	4 acoustic
1	4qPNDBW	Ben Wood	Ghost (Ac	Ghost - Ac	55	149610	FALSE	0.42	0.166	1	-17.235		1	0.0763	0.924	5.56E-06	0.101	0.267	77.489	4 acoustic
2	1uBSr7s7j	Ingrid Micl	To Begin A	To Begin A	57	210826	FALSE	0.438	0.359	0	-9.734		1	0.0557	0.21	0	0.117	0.12	76.332	4 acoustic
3	6lfxq3CG4	Kina Grani	Crazy Rich	Can't Help	71	201933	FALSE	0.266	0.0596	0	-18.515		1	0.0363	0.905	7.07E-05	0.132	0.143	181.74	3 acoustic
4	5vjLSffimil	Chord Ove	Hold On	Hold On	82	198853	FALSE	0.618	0.443	2	-9.681		1	0.0526	0.469	0	0.0829	0.167	119.949	4 acoustic
5	01MVOI9K	Tyrone We	Days I Will	Days I Will	58	214240	FALSE	0.688	0.481	6	-8.807		1	0.105	0.289	0	0.189	0.666	98.017	4 acoustic
6	6Vc5wAM	A Great Bi	Is There A	Say Somet	74	229400	FALSE	0.407	0.147	2	-8.822		1	0.0355	0.857	2.89E-06	0.0913	0.0765	141.284	3 acoustic
7	1EzrEOXm	Jason Mra	We Sing, V	I'm Yours	80	242946	FALSE	0.703	0.444	11	-9.331		1	0.0417	0.559	0	0.0973	0.712	150.96	4 acoustic
8	0lktbUcnA	Jason Mra	We Sing, V	Lucky	74	189613	FALSE	0.625	0.414	0	-8.7		1	0.0369	0.294	0	0.151	0.669	130.088	4 acoustic
9	7k9GuYlY	Ross Copr	Hunger	Hunger	56	205594	FALSE	0.442	0.632	1	-6.77		1	0.0295	0.426	0.00419	0.0735	0.196	78.899	4 acoustic
10	4mzP5mH	Zack Tabu	Episode	Give Me Yc	74	244800	FALSE	0.627	0.363	8	-8.127		1	0.0291	0.279	0	0.0928	0.301	99.905	4 acoustic

First 10 rows of dataset before trimming



STAGE I CONT'D

Feature Exploration/Data Cleaning

Of the initial 20 features that the dataset initially had, we decided to keep **half** of them.

Features such as track_genre contained 'NULL' values in some rows, which were filtered out to prevent unintended data skewing.

With the implementation of a secondary Grammy's dataset, we joined the two datasets on their common 'nominee' and 'winner' features. With the addition of confirmed Grammy winners in the dataset, we can now draw more informed conclusions based on the correlation strength between certain features and a 'winner' attribute evaluating to true.

artists	track_name	popularity	duration_ms	explicit	danceability	valence	tempo	time_signature	track_genre	winner
Gen Hoshi	Comedy	73	230666	FALSE	0.676	0.715	87.917	4	acoustic	FALSE
Ben Wood	Ghost - Ac	55	149610	FALSE	0.42	0.267	77.489	4	acoustic	FALSE
Ingrid Mict	To Begin A	57	210826	FALSE	0.438	0.12	76.332	4	acoustic	FALSE
Kina Granr	Can't Help	71	201933	FALSE	0.266	0.143	181.74	3	acoustic	FALSE
Chord Ove	Hold On	82	198853	FALSE	0.618	0.167	119.949	4	acoustic	FALSE
Tyrone We	Days I Will	58	214240	FALSE	0.688	0.666	98.017	4	acoustic	FALSE
A Great Bi	Say Somet	74	229400	FALSE	0.407	0.0765	141.284	3	acoustic	TRUE
Jason Mra	I'm Yours	80	242946	FALSE	0.703	0.712	150.96	4	acoustic	FALSE
Jason Mra	Lucky	74	189613	FALSE	0.625	0.669	130.088	4	acoustic	TRUE
Ross Copp	Hunger	56	205594	FALSE	0.442	0.196	78.899	4	acoustic	FALSE
Zack Tabur	Give Me Yc	74	244800	FALSE	0.627	0.301	99.905	4	acoustic	FALSE



First 10 rows after
data cleaning

STAGE II REVIEW: STATISTICAL ANALYSIS

Chosen Features: 'tempo',
'duration_ms', and 'winner'



Descriptive Statistics

Many of the features that we chose to keep in our cleaned dataset were categorical rather than numeric, and also could not be encoded into numeric forms that would sufficiently represent them unless they were binary. Because of how powerful of a feature it would be in any of our potential hypotheses, we included the 'winner' feature in our statistical analysis even though it is categorical.

```
The mean tempo is: 122.3081
The mean song duration in milliseconds is: 228090.1387
The average number of award-winning songs is: 0.0278
```

```
The median tempo is: 122.02
The median duration in milliseconds is: 213013.0
The median winning song is: 0.0
```

```
The variance of tempo: 878.8351
The variance of song duration: 11252108517.4830
The variance of award-winning songs: 0.0271
```

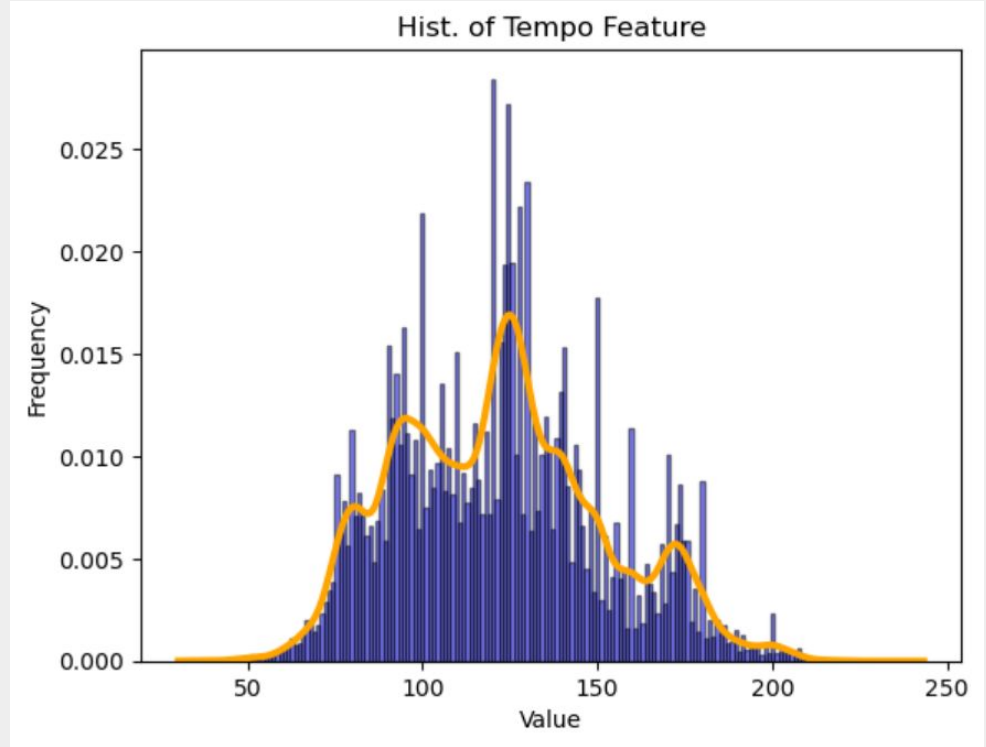
```
The skew of song tempo is: 0.3204
The skew of song duration in seconds is: 10.9618
The skew of winning songs is: 5.7409
```

```
The skew of song tempo is: -0.3786
The skew of song duration in seconds is: 351.7612
The skew of winning songs is: 30.9580
```

STAGE II CONT'D

Distribution Fitting

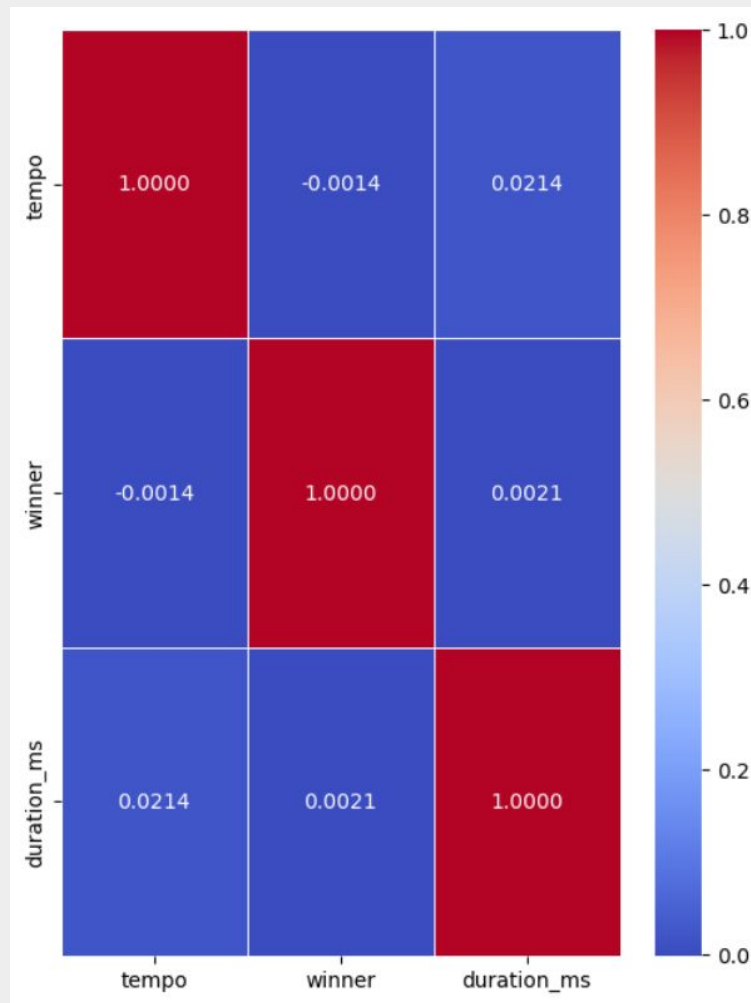
We chose a **Binomial Distribution** for the Tempo Feature based off of its shape. Also because of its only possible outcomes being winner or not of a Grammy, they are best-suited to a Binomial Distribution.



STAGE II CONT'D

Correlation Analysis

There is minimal correlation which implies that musical features are independent of competition outcomes in this dataset. Changes in tempo, or duration of music don't determine Grammy winners.



PROPOSED HYPOTHESES

After completing our statistical analysis, we came up with 3 potential hypotheses that we could propose:

01 Does a higher **popularity** rating lead to higher chances of winning an award?

02 Does **explicitness** of a song lead to lower chances of winning an award?

03 Does a higher **tempo** lead to a higher chance of winning an award?

STAGE III REVIEW: MACHINE LEARNING MODELS

Hypothesis Verification

01

Null: **Popularity** rating has no effect on the chance of winning an award.

Alternative: A higher popularity rating increases the chance of winning an award.

►► For this, we chose the Regression Coefficient Test

02

Null: **Explicitness** of a song has no effect on the chance of winning an award.

Alternative: Explicit songs are less likely to win an award.

►► For this, we chose the Chi-Square Test

03

Null: **Tempo** has no effect on the chance of winning an award.

Alternative: Songs with a higher tempo are more likely to win an award.

►► For this, we chose the Regression Coefficient Test

STAGE III CONT'D

Test Statistic, P-Value, & Significance Level

For each of the respective features and their appropriate test, we evaluated whether or not we failed to reject the null hypothesis based on the test's returned p-value. »

```
Regression Coefficient Test of Popularity & Winner:  
Test Statistic (z): 13.018  
p-value: 0.0000  
Significance Level ( $\alpha$ ): 0.05
```

```
Chi-Square Test of Explicitness & Winner:  
Chi-Square Statistic: 1.865  
p-value: 0.1720  
Significance Level ( $\alpha$ ): 0.05
```

```
Regression Coefficient Test of Tempo & Winner:  
Test Statistic (z): -0.484  
p-value: 0.6287  
Significance Level ( $\alpha$ ): 0.05
```

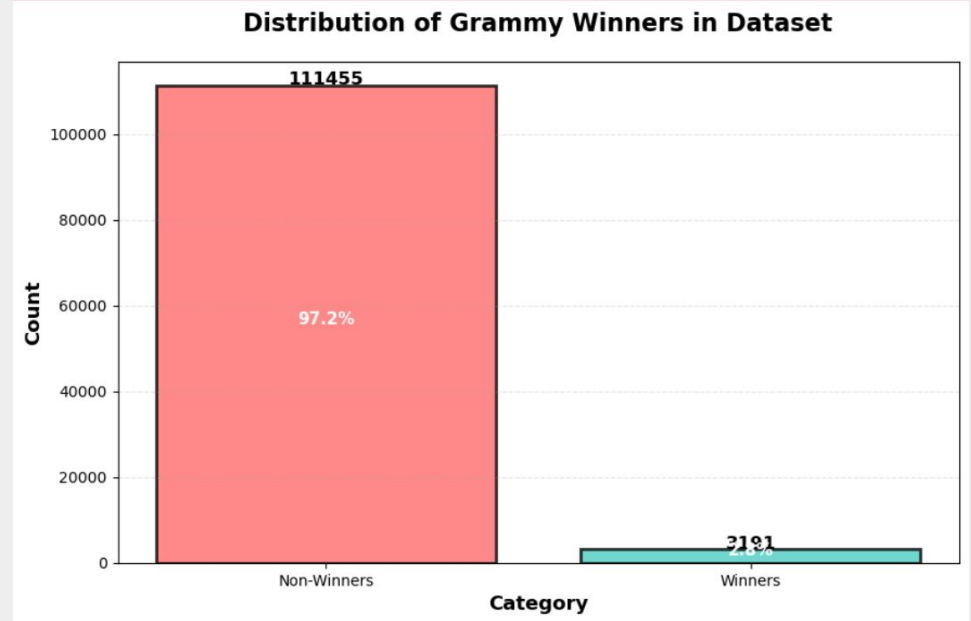
STAGE III CONT'D

Machine Learning Exploration

For the most suitable ML problem type, we chose to use the **classification** problem type (Winner vs Not Winner). When building our model, the binary nature makes for better visualization.

Non-winners (0): 111455 (97.22%)
Winners (1): 3191 (2.78%)

Class Imbalance Ratio: 34.93:1



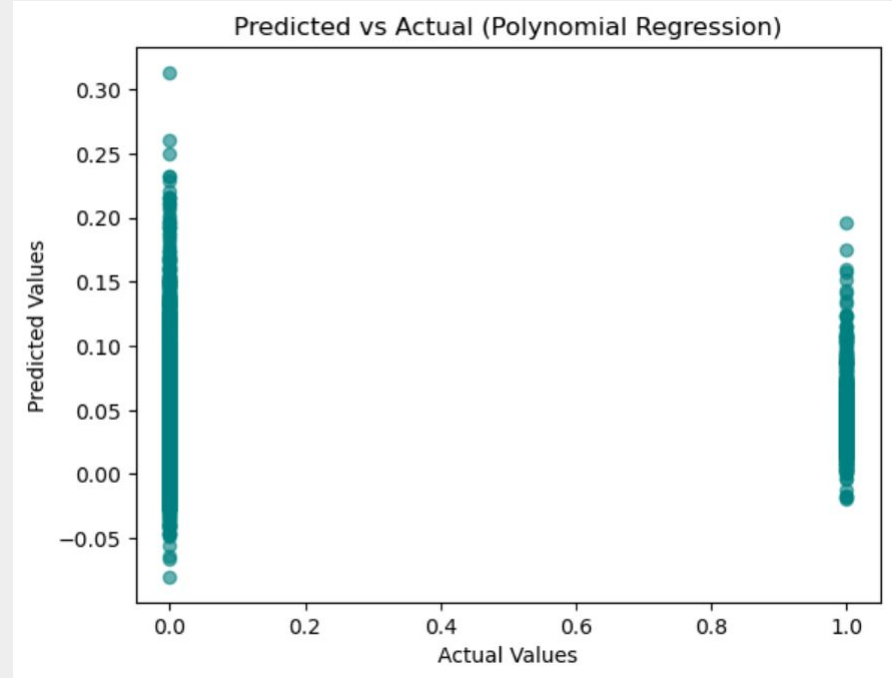
STAGE III CONT'D

Baseline Linear Regression Model

Degree	R2	RMSE	Accuracy	Precision	Recall
1	0.003370	0.165930	0.971566	0.0	0.0
2	0.010555	0.165331	0.971566	0.0	0.0
3	0.011625	0.165241	0.971566	0.0	0.0
5	0.012617	0.165158	0.971566	0.0	0.0

Because our target variable ('winner') is binary, the linear regression model's predictions cluster at 0 and 1, which results in the vertically-separated scatter plot.

For a binary target variable, a linear regression model is not the most informative choice of models.



ADDITIONAL MODELS

Of the additional models, the
Random Forest Classifier
Model performed the best.



Support Vector
Machine



SVM Results:

Accuracy: 0.5612
Precision: 0.0373
Recall: 0.5813

Decision Tree
Classifier



Decision Tree Results:

Accuracy: 0.9727
Precision: 0.5183
Recall: 0.5874

Random Forest
Classifier



Random Forest Classifier Results:

Accuracy: 0.9867
Precision: 1.0000
Recall: 0.5337

Logistic
Regression



Logistic Regression Results:

Accuracy: 0.5614
Precision: 0.0373
Recall: 0.5813

- Highest **Accuracy & Precision**
- But lowest **Recall**

MODEL COMPARISON

	Model	Accuracy	Precision	Recall
0	Polynomial Regression	0.9716	0.0000	0.0000
1	Logistic Regression	0.5614	0.0373	0.5813
2	Support Vector Machine (SVM)	0.5612	0.0373	0.5813
3	Decision Tree Classifier	0.9727	0.5183	0.5874
4	Random Forest Classifier	0.9867	1.0000	0.5337

BEST-PERFORMING MODEL: RANDOM FOREST CLASSIFIER

- This model was able to achieve the highest accuracy of **0.9867**.
- The perfect precision of 1 means every song it predicted as a winner was an **actual winner**.
- The model's **primary drawback** is its recall, only identifying approx. half of the actual winners.

STAGE IV REVIEW: VALIDATION/VISUALIZATION

Model Validation

Cross-Validation Results (Random Forest):
Accuracy: Mean=0.9907, Std=0.0006
F1 Score: Mean=0.7994, Std=0.0157
AUC: Mean=0.8822, Std=0.0059

Reported Mean, Std.
Deviation of metrics after
performing cross
validation with k-folds
method



- With an F1 score of .7994, which is a **significantly high** score ($> .7$), we know that our model is able to **consistently** make precise predictions on our dataset.
- Our AUC also returns a high score of .8822, reflecting its behavior across random thresholds and its ability to distinguish between classes

STAGE IV CONT'D

Model Visualization



CONCLUSIONS

After completing our study on this dataset, we concluded for each of our proposed hypotheses:

01

There is no correlation between popularity and Grammy winners according to our dataset.

02

There is no correlation between explicitness and Grammy winners according to our dataset.

03

There is no correlation between tempo and Grammy winners according to our dataset.